

Low-Complexity Piecewise-Linear Approximations of the Standard Normal CDF for Engineering Calculations: An Accuracy-Complexity Analysis

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github.com/
ramizassaf/normal-cdf-
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Abstract: Engineers still evaluate the standard normal cumulative distribution function $\Phi(z)$ by hand when checking software output, teaching, or working without statistical tools. This paper evaluates piecewise-linear approximations of $\Phi(z)$ on $0 \leq z \leq 3$ as an accuracy-complexity problem. Six three-segment models are compared: exact-secant interpolation through $\Phi(0), \dots, \Phi(3)$; the rounded-coefficient version used in an earlier draft; a continuous two-digit model (0.50, 0.84, 0.98, 1.00); and three models obtained by solving a minimax linear programme with integer, free, and simplicity-constrained breakpoints. All statistics are computed on a 30,001-point grid against a reference CDF verified to 1.1×10^{-16} . Exact-secant interpolation underestimates Φ everywhere inside each segment, as concavity predicts, with maximum absolute error 0.0240 (2.40 percentage points) at $z = 1.47$. Minimax node values at the same integer breakpoints reduce this to 0.0139, and free breakpoints to 0.0092. Counting stored constants as the complexity measure, equal-width breakpoints with minimax node values dominate free breakpoints up to seven constants. Errors of one to two percentage points in Φ translate into relative errors of 35% to 150% in tail probabilities between $z = 1.5$ and $z = 2.8$. These approximations therefore suit rough coverage and yield estimates, and do not suit defect-rate (ppm) estimation.

Keywords: normal distribution; piecewise-linear approximation; minimax approximation; linear programming; accuracy-complexity trade-off; process capability; engineering education

1. Introduction

The standard normal cumulative distribution function (CDF),

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-t^2/2} dt, \quad (1)$$

has no closed form. Statistical software evaluates Φ to machine precision, so the practical case for a hand approximation rests on three narrower needs: checking software output for gross errors, teaching how approximation error behaves, and working in settings without tools. In each case the user trades accuracy for something easier to remember or compute.

Joining tabulated values of Φ by straight lines is classical interpolation and is not new. This paper makes three narrower contributions. First, it separates models that earlier drafts mixed together (exact-secant, rounded, and continuous two-digit models) and states their boundary behaviour. Second, it formulates breakpoint and node-value selection as a minimax problem, solves it, and reports an accuracy-complexity frontier with an explicit complexity measure. Third, it quantifies how CDF errors propagate into the tail probabilities that drive defect-rate estimates.

The analysis is restricted to $0 \leq z \leq 3$. Negative arguments follow from symmetry, $\Phi(-z) = 1 - \Phi(z)$. Applications with $z > 3$ or with small tail probabilities need exact evaluation.

2. Background

Abramowitz and Stegun [1] collect rational and polynomial approximations to Φ , including formula 26.2.17 with absolute error below 7.5×10^{-8} . Hastings [2] developed many of these approximations for early digital computers. Pólya [3] gave the one-line formula $\Phi(z) \approx \frac{1}{2} + \frac{1}{2}\sqrt{1 - \exp(-2z^2/\pi)}$. Bowling et al. [4], writing for an industrial engineering audience, proposed a logistic approximation with maximum error below 1.4×10^{-4} . Modern libraries use algorithms of the Cody or Cephes family for Φ , and Wichura [5] for its inverse.

These formulas require an exponential and sometimes a square root, so they suit a calculator but not mental arithmetic. Piecewise-linear models need only multiplication and addition. Their error follows the classical bound for linear interpolation [6]: on a segment of width h ,

$$|e(z)| \leq \frac{h^2}{8} \max |\Phi''(z)|. \quad (2)$$

With $\Phi''(z) = -z\phi(z)$, whose magnitude peaks at $z = 1$ with value $\phi(1) = 0.2420$, the bound for $h = 1$ is 0.0302.

Applications in quality engineering use Φ for process yield and capability [7] and for statistical tolerance analysis [8].

3. Notation and Error Metrics

Let $\hat{\Phi}(z)$ denote an approximation. All errors are stated in probability units unless marked otherwise.

- Signed error: $e(z) = \hat{\Phi}(z) - \Phi(z)$. Negative values mean underestimation.
- Absolute error: $|e(z)|$. A value of 0.0240 equals 2.40 percentage points (pp); the conversion is $100 |e|$ pp.
- Relative CDF error: $100 |e(z)| / \Phi(z)$, in percent.
- Tail probability $Q(z) = 1 - \Phi(z)$ and its relative error $r_Q(z) = 100 (\hat{Q} - Q) / Q$, in percent, with $\hat{Q} = 1 - \hat{\Phi}$.
- Two-sided coverage of a centred symmetric interval: $P(|Z| \leq z) = 2\Phi(z) - 1$. Its error is $2e(z)$, double the CDF error.

Statistics are evaluated on $z = 0, 0.0001, \dots, 3$ (30,001 points). The reference Φ is `scipy.special.ndtr` [9], checked against 50-digit `mpmath` [10] at 3,001 points; the largest difference is 1.1×10^{-16} .

4. Three Interpolation-Based Models

4.1. Model A: exact secant

Model A joins $(k, \Phi(k))$, $k = 0, 1, 2, 3$, with straight lines, using unrounded values (Table 1):

$$\hat{\Phi}_A(z) = \Phi(k) + m_k(z - k), \quad k \leq z \leq k + 1, \quad m_k = \Phi(k + 1) - \Phi(k). \quad (3)$$

Model A is continuous by construction.

Table 1. Breakpoint values and exact secant slopes used by Model A.

k	$\Phi(k)$	Segment	Slope m_k
0	0.500000	[0,1]	0.341345
1	0.841345	[1,2]	0.135905
2	0.977250	[2,3]	0.021400
3	0.998650	–	–

Sign of the error. For $z > 0$, $\Phi''(z) = -z\phi(z) < 0$, so Φ is strictly concave on the whole positive half-axis. A chord of a strictly concave function lies below the function between its end points. Hence $\hat{\Phi}_A(z) < \Phi(z)$ inside every segment and $e_A(z) = 0$ only at the breakpoints. Model A underestimates Φ ; its errors do not oscillate around zero. The computation confirms this: the largest positive error of Model A is 0.0000 (Table 5).

4.2. Model B: rounded coefficients, as originally published

The earlier draft rounded the slopes to two decimals and kept three-decimal intercepts:

$$\hat{\Phi}_B(z) = \begin{cases} 0.50 + 0.34z, & 0 \leq z \leq 1, \\ 0.841 + 0.14(z - 1), & 1 < z \leq 2, \\ 0.977 + 0.02(z - 2), & 2 < z \leq 3. \end{cases} \quad (4)$$

Model B is discontinuous. With the closed-right convention above, Table 2 lists the behaviour at each interior breakpoint. The jump at $z = 2$ is -0.004 , and just left of $z = 2$ Model B overestimates Φ by $+0.0038$, so rounding breaks the one-sided error property of Model A. We report Model B for completeness and do not recommend it.

Table 2. Boundary behaviour of Model B at its interior breakpoints.

z	$\Phi(z)$	$\hat{\Phi}_B(z)$	Right limit	Jump	$e_B(z)$
1	0.8413	0.840	0.841	+0.001	-0.0013
2	0.9772	0.981	0.977	-0.004	+0.0038

4.3. Model C: continuous two-digit nodes (“50-84-98-100”)

Model C keeps the slopes 0.34, 0.14 and 0.02 but chains the intercepts so the function is continuous:

$$\hat{\Phi}_C(z) = \begin{cases} 0.50 + 0.34z, & 0 \leq z \leq 1, \\ 0.84 + 0.14(z - 1), & 1 < z \leq 2, \\ 0.98 + 0.02(z - 2), & 2 < z \leq 3. \end{cases} \quad (5)$$

The four node values 0.50, 0.84, 0.98 and 1.00 are easy to recall because $\Phi(1)$ and $\Phi(2)$ follow from the 68-95 rule: $\Phi(1) = (1 + 0.6827)/2$ and $\Phi(2) = (1 + 0.9545)/2$. Model C overestimates near $z = 2$ and at $z = 3$, where $\hat{\Phi}_C(3) = 1.00$ forces the tail estimate to zero.

Table 3. Values and signed errors of Models A, B and C at selected points.

z	$\Phi(z)$	$\hat{\Phi}_A$	e_A	$\hat{\Phi}_B$	e_B	$\hat{\Phi}_C$	e_C
0.0	0.5000	0.5000	0.0000	0.5000	0.0000	0.5000	0.0000
0.5	0.6915	0.6707	-0.0208	0.6700	-0.0215	0.6700	-0.0215
1.0	0.8413	0.8413	0.0000	0.8400	-0.0013	0.8400	-0.0013
1.5	0.9332	0.9093	-0.0239	0.9110	-0.0222	0.9100	-0.0232
2.0	0.9772	0.9772	0.0000	0.9810	+0.0038	0.9800	+0.0028
2.5	0.9938	0.9879	-0.0058	0.9870	-0.0068	0.9900	-0.0038
3.0	0.9987	0.9987	0.0000	0.9970	-0.0017	1.0000	+0.0013

Table 3 shows the trap in checking only half-integer points: Model A’s largest error on $[0, 3]$ occurs at $z = 1.47$ (-0.0240), not at $z = 0.5$ (-0.0208). The local maximum on the first segment is -0.0211 at $z = 0.56$. All three values sit below the bound of Equation (2).

5. Optimisation Formulation

For k segments with breakpoints $0 = b_0 < b_1 < \dots < b_k = 3$ and node values $y_0, \dots, y_k$, let $L(z; \mathbf{b}, \mathbf{y})$ be the continuous piecewise-linear function through (b_j, y_j) . The design problem is

$$\min_{\mathbf{b}, \mathbf{y}} \max_{0 \leq z \leq 3} |L(z; \mathbf{b}, \mathbf{y}) - \Phi(z)| \quad \text{s.t.} \quad y_0 = 0.5, \quad 0 \leq y_j \leq 1. \quad (6)$$

The constraint $y_0 = 0.5$ keeps the symmetric extension $\hat{\Phi}(-z) = 1 - \hat{\Phi}(z)$ continuous at zero, and the bounds keep $\hat{\Phi}$ a valid probability. Continuity is built into L . Monotonicity was not imposed; every solution reported below satisfies it.

For fixed $\mathbf{b}$, problem (6) on the grid is a linear programme in $(\mathbf{y}, t)$: minimise t subject to $-t \leq L(z_i) - \Phi(z_i) \leq t$ at every grid point. We solve it with the HiGHS solver [11] through SciPy. Free breakpoints are found by an outer Nelder-Mead search [12] with seven starting points. The outer search is local, so free-breakpoint results are the best found, not proven global optima. Three solutions are reported:

- **Model D:** integer breakpoints, minimax node values (the linear programme alone).
- **Model E:** free breakpoints, minimax node values.

- **Model F:** simplicity-constrained optimum. Breakpoints are restricted to multiples of 0.5 and node values to two decimals; all combinations within ± 0.02 of the continuous optimum are searched exhaustively.

Table 4. Optimised three-segment models (breakpoints and node values rounded to four decimals for display; full precision in `results.json`).

Model	Breakpoints $\mathbf{b}$	Node values $\mathbf{y}$	Max $ e $
D	(0, 1, 2, 3)	(0.5000, 0.8552, 0.9834, 0.9848)	0.0139
E	(0, 0.8597, 1.6746, 3)	(0.5000, 0.8143, 0.9622, 1.0000)	0.0092
F	(0, 1, 2, 3)	(0.5000, 0.8500, 0.9900, 0.9900)	0.0165

Model F lands on integer breakpoints with node values 0.50, 0.85, 0.99, 0.99. Its last segment is flat, which minimises the CDF error but gives a poor tail (Section 6.4). Rounding Model E's constants to four decimals raises its maximum error from 0.00923 to 0.00928.

6. Results

6.1. Error statistics

Table 5. Error statistics on $0 \leq z \leq 3$ (30,001 points). Columns 2 to 8 in probability units; multiply by 100 for percentage points.

Model	Max $ e $	at z	Mean e	Mean $ e $	RMSE	95th pct $ e $	Max $e > 0$	Max rel. (%)
A: exact secant	0.0240	1.47	-0.0112	0.0112	0.0136	0.0234	0.0000	3.01
B: rounded (published)	0.0225	1.45	-0.0111	0.0112	0.0133	0.0219	+0.0038	3.10
C: 50–84–98–100	0.0235	1.45	-0.0105	0.0106	0.0134	0.0229	+0.0028	3.10
D: minimax, integer b	0.0139	1.51	-0.0068	0.0085	0.0095	0.0138	+0.0139	2.05
E: minimax, free b	0.0092	2.30	-0.0041	0.0057	0.0064	0.0092	+0.0092	1.41
F: two-decimal optimum	0.0165	0.51	-0.0055	0.0079	0.0091	0.0160	+0.0128	2.40

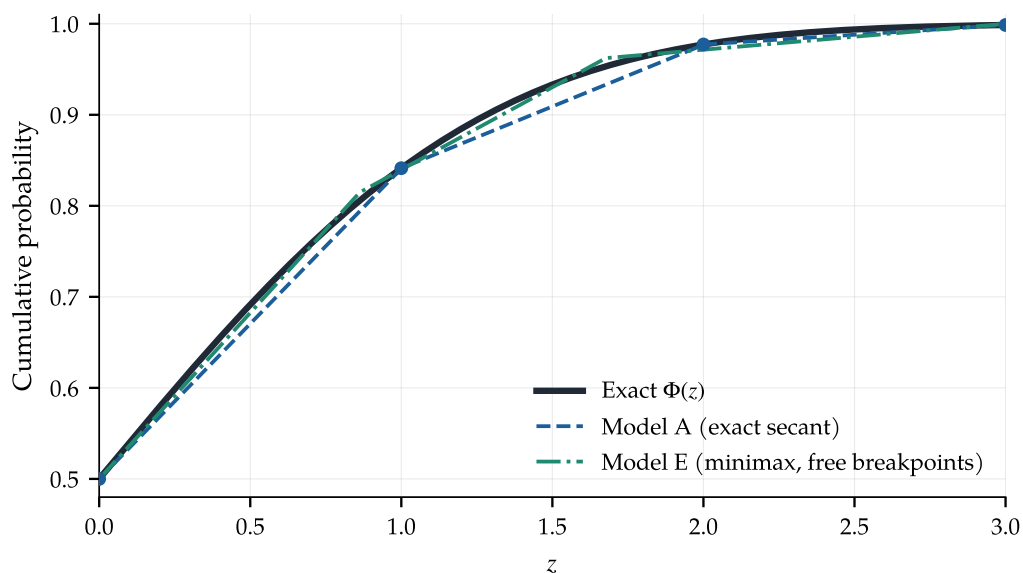


Figure 1. Exact $\Phi(z)$, Model A (exact secant through the dots) and Model E (minimax with free breakpoints).

Three results stand out in Table 5 and Figure 2. Model A's mean signed error equals minus its mean absolute error, confirming one-sided underestimation; earlier drafts reported a mean error near zero, which was incorrect. Rounding (Models B and C) changes the maximum error by less

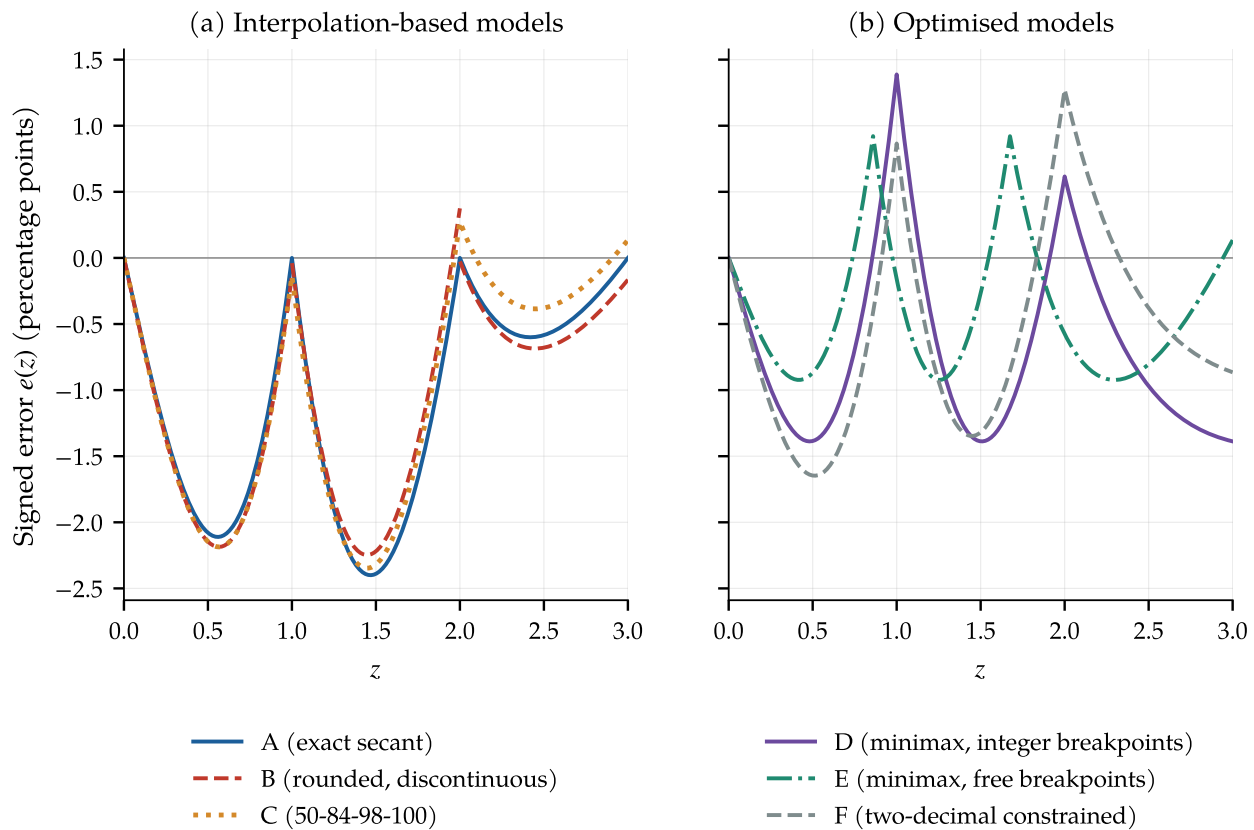


Figure 2. Signed error $e(z)$ in percentage points. (a) Interpolation-based models: Model A stays at or below zero, consistent with concavity; Model B jumps at $z = 1$ and $z = 2$. (b) Optimised models equalise positive and negative peaks, which is the signature of a minimax solution.

than 0.002 but introduces overestimation. Choosing node values by minimax (Model D) cuts the maximum error by 42% with the same integer breakpoints and the same arithmetic effort.

6.2. Accuracy-complexity frontier

We measure complexity as the number of constants a user must store. Equal-width models store $k + 1$ node values (breakpoints are implied); free-breakpoint models also store $k - 1$ interior breakpoints, for $2k$ constants in total. Table 6 and Figure 3 report the best maximum error found for $k = 1, \dots, 6$.

Table 6. Maximum absolute error by number of segments k and number of stored constants.

k	Equal-width breakpoints			Free breakpoints	
	Constants	Interpolation	Minimax values	Constants	Minimax values
1	2	0.1872	0.1866	2	0.1866
2	3	0.0571	0.0381	4	0.0244
3	4	0.0240	0.0139	6	0.0092
4	5	0.0164	0.0082	8	0.0048
5	6	0.0106	0.0053	10	0.0030
6	7	0.0071	0.0035	12	0.0025

Two conclusions follow, both conditional on this complexity measure. Per segment, free breakpoints give the smallest error. Per stored constant, equal-width breakpoints with minimax node values dominate: four constants give 0.0139, while free breakpoints need six constants to reach 0.0092 and eight to reach 0.0048. The earlier draft claimed that six equal-width interpolation segments give a maximum error near 0.0005; the computed value is 0.0071, and that claim is

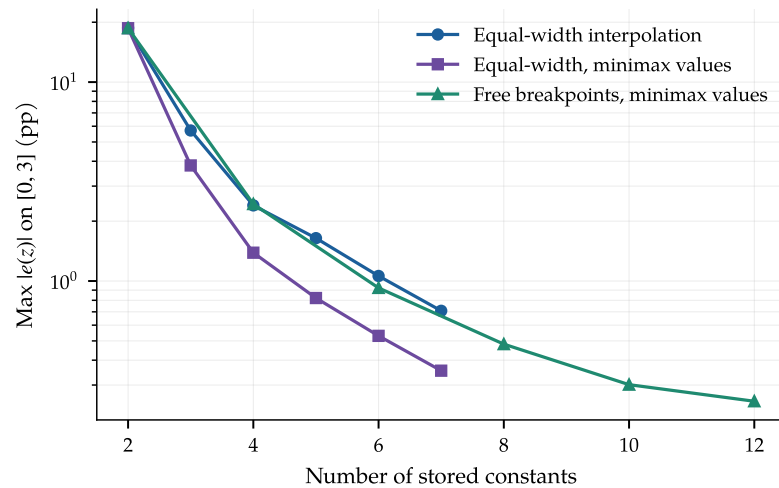


Figure 3. Accuracy-complexity frontier. For a fixed number of stored constants, equal-width breakpoints with minimax node values give the smallest error in this study.

withdrawn.

The frontier is a frontier of the models computed here. Human effort also depends on the arithmetic (multiplying by 0.34 versus 0.3655), which the constant count does not capture. A study with users would be needed to measure that cost.

6.3. Reference non-linear approximations

Table 7 places the linear models in context. Pólya's formula reaches 0.0031 with a single expression, an order of magnitude better than any three-segment linear model, at the cost of an exponential and a square root.

Table 7. Classical closed-form approximations evaluated on the same grid.

Approximation	Max $ e $	at z	Operations
Pólya (1949) [3]	3.15×10^{-3}	1.65	1 exp, 1 sqrt
Bowling et al. (2009) [4]	1.41×10^{-4}	1.48	1 exp, 1 division
Abramowitz–Stegun 26.2.17 [1]	7.45×10^{-8}	0.72	1 exp, 6 constants

6.4. Tail probabilities

Defect rates depend on $Q(z) = 1 - \Phi(z)$, which is small exactly where the CDF error looks small. Table 8 reports two-sided out-of-specification rates, $2Q(z) \times 10^6$ ppm, for a centred process.

Table 8. Two-sided defect rates (ppm) and relative tail error r_Q (%) for a centred process.

z	Exact ppm	Model A ppm	r_Q A	Model C ppm	r_Q C	Model E ppm	r_Q E
1.0	317,311	317,311	+0.0	320,000	+0.8	320,531	+1.0
1.5	133,614	181,405	+35.8	180,000	+34.7	138,978	+4.0
2.0	45,500	45,500	+0.0	40,000	-12.1	57,014	+25.3
2.5	12,419	24,100	+94.1	20,000	+61.0	28,507	+129.5
3.0	2,700	2,700	+0.0	0	-100.0	0	-100.0

At $z = 2.5$, Model A's CDF error is only -0.0058 , yet it reports 24,100 ppm instead of 12,419 ppm, an overestimate of 94%. The minimax models do not fix this: Model E is optimal for the CDF and still overestimates the tail at $z = 2.5$ by 130%. A model meant for defect rates needs a different objective, for example minimax relative error in Q .

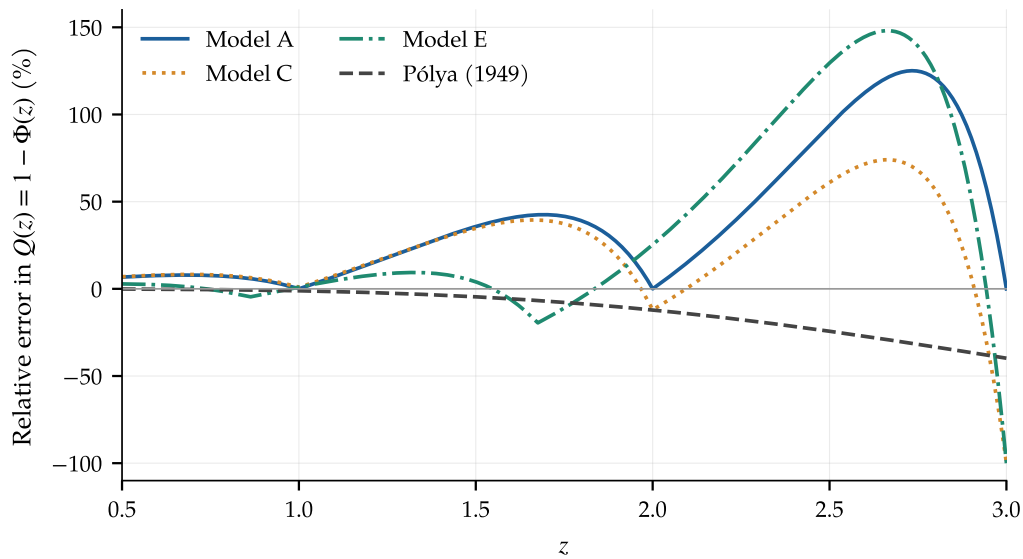


Figure 4. Relative error in the tail probability $Q(z)$. Errors below 2.5 percentage points in Φ become tail errors of 35% to 150%. Model C and Model E return $\hat{Q}(3) = 0$, a relative error of -100% .

7. Engineering Examples

7.1. Centred process yield

A bearing diameter has specification limits ± 0.050 mm around nominal, process standard deviation $\sigma = 0.020$ mm, and a centred mean. The distance to each limit is $z = 0.050/0.020 = 2.5$, so $C_p = 0.100/(6 \times 0.020) = 0.833$. Two-sided yield is $2\Phi(2.5) - 1$.

- Exact: $\Phi(2.5) = 0.99379$; yield 98.758%; 12,419 ppm out of specification.
- Model A: $\hat{\Phi}_A(2.5) = 0.98795$; yield 97.590% (-1.17 pp); 24,100 ppm (+94%).
- Model C: $\hat{\Phi}_C(2.5) = 0.99$; yield 98.000% (-0.76 pp); 20,000 ppm (+61%).

The yield estimates are within about one percentage point. The ppm estimates are off by a factor between 1.6 and 1.9.

7.2. One-sided versus two-sided probability

The probability of conformance to a bilateral tolerance $\pm T$ with stack-up standard deviation σ is $P(|Z| \leq T/\sigma) = 2\Phi(T/\sigma) - 1$, not $\Phi(T/\sigma)$. For $T/\sigma = 1.2$, the exact value is $2(0.88493) - 1 = 0.76986$. Model C gives $\hat{\Phi}_C(1.2) = 0.84 + 0.14(0.2) = 0.868$ and a coverage of 0.736, an error of -3.39 pp, double its CDF error of -1.69 pp.

7.3. Outside the range

A centred process with $C_{pk} = 1.33$ has $z = 3C_{pk} \approx 4$. The exact two-sided yield is 99.9937% (63 ppm). This lies outside $[0, 3]$ and none of the models here applies.

8. Quick Field Rules

For use without tables, a calculator, or software, Model C reduces to four remembered numbers and three slopes.

1. Remember the nodes 0.50, 0.84, 0.98 at $z = 0, 1, 2$ (from the 68-95 rule) and 1.00 at $z = 3$.
2. Find the segment containing $|z|$ and add slope $\times$ distance: 0.34 on $[0, 1]$, 0.14 on $[1, 2]$, 0.02 on $[2, 3]$.
3. For negative z , use $1 - \hat{\Phi}(|z|)$. For two-sided coverage, use $2\hat{\Phi} - 1$.

Corrected worked examples (exact values in parentheses): $z = 0.5$ gives 0.670 (0.6915, -2.15 pp); $z = 1.5$ gives 0.910 (0.9332, -2.32 pp); $z = 1.8$ gives 0.952 (0.9641, -1.21 pp); $z = 2.5$ gives 0.990 (0.9938, -0.38 pp).

When a calculator is available, Pólya's formula reduces the maximum error to 0.0031 with one line and no table.

Suitability. Model C keeps $|e| \leq 0.0235$ on $[0, 3]$, so two-sided coverage stays within ± 4.7 pp. These rules suit decisions that do not change within that band, such as rough yield screening, sanity checks of software output, and classroom estimation. They do not suit ppm or defect-rate estimates, capability claims near specification limits, safety-critical reliability, or any $z > 3$.

Table 9 lists exact reference values for common z values, for a centred process without the 1.5σ shift convention of Six Sigma practice. This table corrects the earlier draft, in which four of the six ppm entries were wrong.

Table 9. Exact reference values for a centred process (no 1.5σ shift).

z	$\Phi(z)$	Two-sided coverage (%)	Two-sided ppm	One-sided ppm
1.0	0.84134	68.269	317,311	158,655
1.5	0.93319	86.639	133,614	66,807
2.0	0.97725	95.450	45,500	22,750
2.5	0.99379	98.758	12,419	6,210
3.0	0.99865	99.730	2,700	1,350
3.5	0.99977	99.953	465	233
4.0	0.99997	99.994	63	32

9. Implementation

A spreadsheet version of Model C, symmetric for negative z and returning #N/A outside $[-3, 3]$ (cell A2 holds z):

```
=IF(ABS(A2)>3,NA(),0.5+SIGN(A2)*IF(ABS(A2)<=1,0.34*ABS(A2),
IF(ABS(A2)<=2,0.34+0.14*(ABS(A2)-1),0.48+0.02*(ABS(A2)-2))))
```

Enter the formula as a single line; a copy is in `spreadsheet/model_C_formula.txt` in the repository. The formula reproduces Model C and its symmetric extension to within 1.1×10^{-16} on 601 test points in $[-3, 3]$. Returning #N/A outside the range is deliberate: silent extrapolation would produce probabilities above one. The repository also contains Python reference code and an interactive browser application that plots all models, their errors and the tail effect, and computes yield and ppm for user-supplied specification limits.

10. Limitations

The minimax problem is solved on a discrete grid with step 10^{-4} ; the continuous optimum can differ slightly. Free breakpoints come from a local search with multiple starts, so better solutions may exist. The complexity measure counts constants and ignores arithmetic difficulty and memory errors; no user study was performed. All objectives concern absolute CDF error on $[0, 3]$; a tail-weighted or relative-error objective would produce different models and is left for future work.

11. Conclusions

Piecewise-linear approximation of $\Phi(z)$ on $[0, 3]$ is a classical idea; its value lies in a clear statement of what it can and cannot do. Exact-secant interpolation through integer points underestimates Φ with a maximum error of 0.0240 at $z = 1.47$. Choosing node values by minimax at the same breakpoints reduces this to 0.0139 at no extra effort, and free breakpoints reach 0.0092. Measured by stored constants, equal-width breakpoints with minimax node values give the best trade-off among the models computed. The continuous two-digit Model C (0.50, 0.84, 0.98, 1.00) is the most memorable option, with a maximum error of 0.0235. For all linear models, errors of one to

two percentage points in Φ become errors of tens of percent in tail probabilities, so these tools serve rough yield and coverage estimates and should not replace exact computation of defect rates.

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Data Availability Statement: All code, the full error grid and every reported statistic are available at <https://github.com/ramizassaf/normal-cdf-piecewise-linear>. Running `analysis/compute_results.py` regenerates all numbers in this paper.

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Conflicts of Interest: The author declares no conflicts of interest.

Revision note: Version 1 of this white paper contained numerical errors, including the location and size of the maximum error, the boundary values of Model B, the defect-rate table, a six-segment error claim, and validation statistics that had not been computed. Version 2 replaces all of them with values generated by the published code.

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